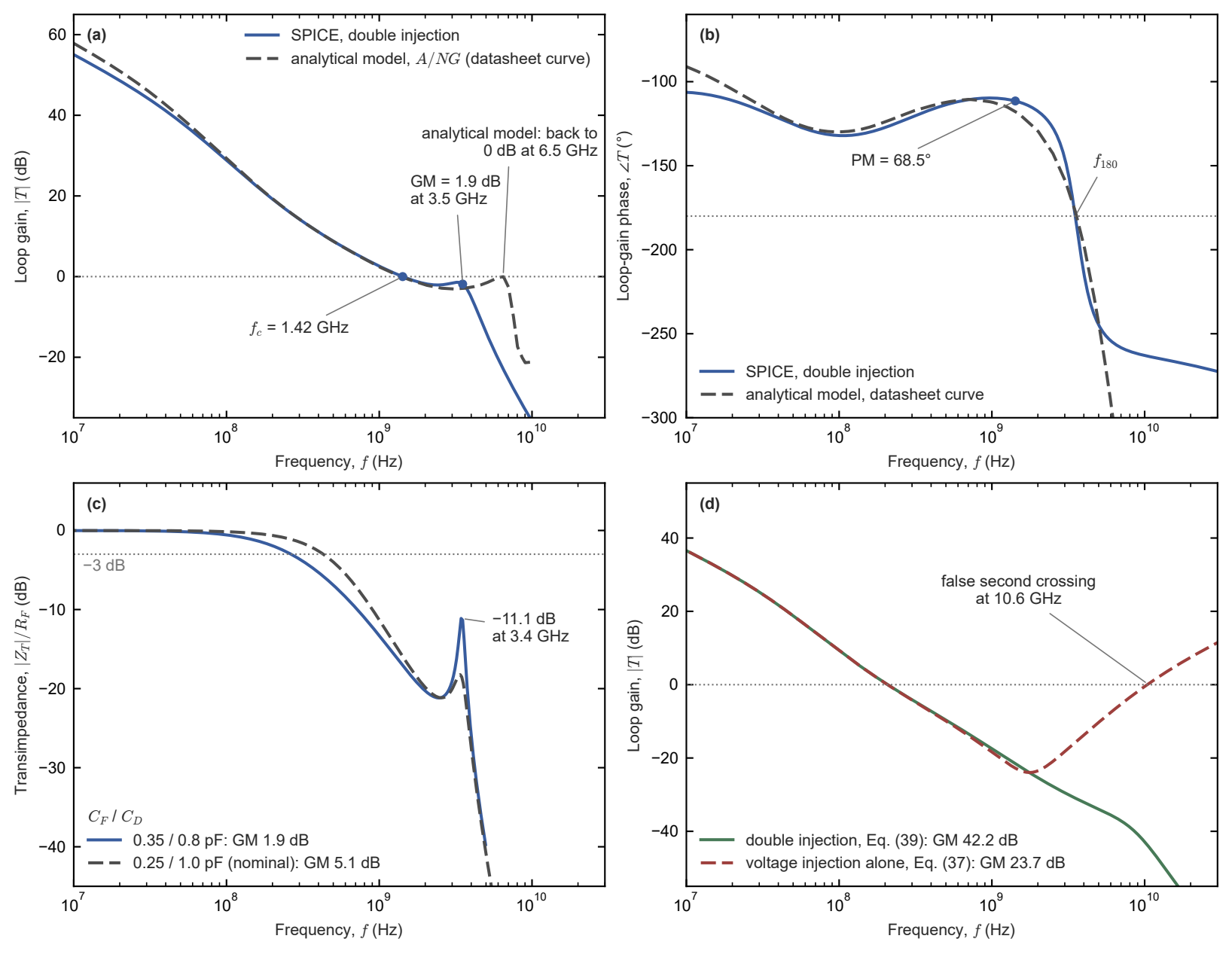

(a)
SPICE, double injection
analytical model, A/NG (datasheet curve)
analytical model: back to 0 dB at 6.5 GHz
GM = 1.9 dB at 3.5 GHz
f_c = 1.42 GHz
Loop gain, $|T|$ (dB)
Frequency, f (Hz)
(b)
PM = 68.5°
f_{180}
SPICE, double injection
analytical model, datasheet curve
Loop-gain phase, $\angle T$ (°)
Frequency, f (Hz)
(c)
−3 dB
−11.1 dB at 3.4 GHz
C_F / C_D
0.35 / 0.8 pF: GM 1.9 dB
0.25 / 1.0 pF (nominal): GM 5.1 dB
Transimpedance, $|Z_T|/R_F$ (dB)
Frequency, f (Hz)
(d)
false second crossing at 10.6 GHz
double injection, Eq. (39): GM 42.2 dB
voltage injection alone, Eq. (37): GM 23.7 dB
Loop gain, $|T|$ (dB)
Frequency, f (Hz)